Final Simulation Design and Algorithms

Product-Mixture Probit Factors versus Viroli-Style Gibbs Baselines

1 Goal

The purpose of this simulation is to stress-test computation and parameter recovery in high-dimensional binary factor mixture models. The comparison is designed around the regime that motivates the IFEval analysis: small to moderate numbers of systems, many binary items, sparse but overlapping item loadings, non-Gaussian factor coordinates, and item intercepts.

The central question is whether the product-mixture MAP estimator can recover the factor scores, loadings, intercepts, and marginal mixture parameters more accurately or more quickly than full Gibbs sampling baselines that fit the same latent factor structure.

2 Data-Generating Model

For each replication and each simulation setting, we generate binary data from

$$\eta_{ih} \mid c_{ih} = g \sim N(\mu_{hg}, \sigma^2_{hg}), \qquad c_{ih} \sim \text{Categorical}(\pi_{h1}, \dots, \pi_{hG_h}), \tag{1}$$

$$Z_{ij} \mid \boldsymbol{\eta}_i \sim N(\alpha_j + \boldsymbol{\lambda}_j^\top \boldsymbol{\eta}_i, 1), \tag{2}$$

$$X_{ij} = \mathbf{1}\{Z_{ij} > 0\}. \tag{3}$$

The factor coordinates are independent in the data-generating distribution, so the joint factor density has product structure

$$p(\boldsymbol{\eta}_i) = \prod_{h=1}^{H} \sum_{g=1}^{G_h} \pi_{hg}\, \phi(\eta_{ih}; \mu_{hg}, \sigma^2_{hg}).$$

All methods are evaluated on exactly the same simulated data for a given setting and replication.

3 Simulation Grid

The final grid is

$$n \in \{100, 200, 300, 400\}, \quad p \in \{500, 1000, 2000, 4000\}, \quad H \in \{5, 10, 15, 20\},$$

$$G \in \{2, 3\}, \quad \text{separation} \in \{1, 2\}, \quad \text{replications} = 25.$$

Unless overridden, the loading design is the IFEval-like "Cross" design: item blocks are moderately unbalanced and items primarily load on one factor while also having sparse cross-loadings. Intercepts are included using the IFEval-like block intercept pattern.

To manage runtime, the product MAP estimator is run over the full p grid by default. The Viroli Gibbs baselines are run for the smaller p grid specified by `P_VALUES_VIROLI`; the default is $p \in \{500, 1000\}$. This is an explicit runtime policy, not a change in the estimand.

Quantity	Default in final launcher
Script	`scripts/sample_size/run_final_product_viroli_simulation.R`
Core driver	`scripts/sample_size/compare_original_simulation_joint_mfa_gibbs.R`
DGP utilities	`R/sample_size_dgp.R`
Loading design	`balanced_moderate_dense_signed_cross`
Block sizes	`moderate_ifeval_like`
Mixture parameters	`viroli_smoke`
Product MAP penalty	`LAMBDA_L1_PENALTY=10`
Pretraining loading penalty	`PRETRAIN_LOADING_PENALTY=10`
Rotation loading penalty	`ROTATION_LOADING_L1_PENALTY=10`
Viroli Laplace penalty	`VIROLI_LAPLACE_L1_PENALTY=10`
Gibbs iterations	2000 total, 1000 burn-in, thinning 1

4 Method 1: Product-Mixture MAP

The product MAP method has three stages: estimate the low-rank probit signal, rotate the estimated score space toward independent marginal mixtures, and refine all parameters by MAP.

4.1 Stage 1: Low-Rank Probit Signal

Instead of sampling an augmented matrix Z, the first stage estimates the rank-H signal matrix B in the probit likelihood

$$\ell(\boldsymbol{\alpha}, B) = \sum_{i=1}^{n} \sum_{j=1}^{p} \left[X_{ij} \log \Phi(\alpha_j + B_{ij}) + (1 - X_{ij}) \log\{1 - \Phi(\alpha_j + B_{ij})\}\right], \qquad \operatorname{rank}(B) \le H.$$

This is a direct likelihood target for the latent linear predictor $B = F\Lambda^\top$. The implementation uses a simple EM-SVD routine:

1. Given the current $\boldsymbol{\alpha}$ and B, compute the conditional expectation of the truncated Gaussian latent variable Z_{ij} under the binary probit observation.
2. Update $\boldsymbol{\alpha}$ and project the centered conditional expectation matrix onto rank H using an SVD.
3. Stop when the probit log likelihood or the low-rank signal changes by less than the tolerance, or when the maximum iteration count is reached.

   The output is $\widehat{B}$. Its SVD gives

$$\widehat{B} = \widehat{U}\widehat{D}\widehat{V}^\top, \qquad \widehat{S} = \sqrt{n}\,\widehat{U}, \qquad \widehat{\Lambda}_{\text{svd}} = \widehat{V}\widehat{D}/\sqrt{n}.$$

Under standard low-rank probit regularity conditions, consistency of $\widehat{B}$ implies consistency of the estimated signal subspace, which provides a principled initialization for the left singular vector space. This stage is the main computational improvement over sampled-Z pretraining: it targets the binary likelihood directly and avoids repeated augmentation draws.

4.2 Stage 2: Rotation

The SVD identifies the rank-H subspace but not the independent coordinate axes. The rotation stage searches over orthogonal matrices R and evaluates

$$Q(R) = \sum_{h=1}^{H} \max_{q_h \in \mathcal{M}_{G_h}} \frac{1}{n} \sum_{i=1}^{n} \log q_h\{(\widehat{S}R)_{ih}\} - \lambda_{\mathrm{rot}} \|\widehat{\Lambda}_{\mathrm{svd}} R\|_1.$$

The first term rewards rotated coordinates that are well described by one-dimensional Gaussian mixtures. The second term favors sparse rotated loadings. The implementation performs pairwise Givens sweeps, fitting the marginal mixtures by EM/MAP after candidate rotations. The rotation stage is terminated only after the rotation objective is stable and, by default, all inner marginal mixtures have converged.

4.3 Stage 3: MAP Refinement

Starting from the rotated scores and loadings, the refinement stage alternates:

1. Update factor scores by MAP under the probit likelihood and current marginal mixture prior.
2. Update item intercepts and sparse loadings by itemwise penalized probit regressions, using an ℓ_1 penalty on loadings.
3. Update the marginal mixture means, variances, and weights by MAP.

The refinement objective is monitored after each full sweep. The monotonicity guard rejects a sweep if the selected posterior objective decreases beyond a small tolerance. The stage stops when the posterior objective is stable after the minimum number of iterations and, by default, the marginal mixture fits have also converged.

5 Method 2: Viroli-Style Gibbs with Laplace Loadings

This baseline fits the correctly specified independent marginal mixture factor model by Gibbs sampling with probit augmentation. Each iteration updates:

1. Z_{ij} from its truncated normal full conditional given X_{ij}, α_j, $\boldsymbol{\lambda}_j$, and $\boldsymbol{\eta}_i$.
2. The factor scores $\boldsymbol{\eta}_i$ and coordinate-wise component labels c_{ih}.
3. Marginal mixture weights, means, and variances for each factor coordinate.
4. Item intercepts and loadings.

The Laplace loading prior is implemented through the Bayesian lasso scale mixture representation. Posterior draws are normalized to the convention that each factor coordinate has marginal mixture mean zero and variance one, with the corresponding inverse transformation applied to $\boldsymbol{\alpha}$ and Λ so that the probit linear predictor is unchanged. The sampler keeps 1000 posterior draws after 1000 burn-in iterations.

6 Method 3: Viroli-Style Gibbs with Diffuse Gaussian Loadings

The Gaussian Viroli baseline uses the same probit augmentation and independent marginal mixture factor prior as the Laplace version, but replaces the Bayesian lasso loading prior with a diffuse

Gaussian loading prior. This isolates how much of the Gibbs performance comes from the sampling scheme itself versus the sparsity prior on Λ.

7 Alignment and Metrics

All methods are aligned to the data-generating factors using the same permutation and sign strategy. The final launcher uses loading-based alignment by default: it chooses the permutation and sign pattern that best matches the data-generating loading matrix, then applies the same transformation to factor scores and mixture parameters. Factor-score alignment can be requested with `ALIGNMENT_MODE=factors`. After alignment, the reported metrics are:

- factor-score RMSE, computed after standardizing true and estimated scores;
- loading RMSE and loading correlation;
- item-intercept RMSE and correlation;
- marginal mixture mean, variance, and weight RMSE;
- predictive probability RMSE;
- elapsed seconds per replication;
- effective sample size summaries for Gibbs posterior draws;
- DGP diagnostics, including item prevalence summaries, empty marginal components, and the number of observed joint profiles.

When G^H is small enough, the driver also writes joint mixture recovery tables. For large H, these dense joint tables are intentionally skipped and the marginal mixture summaries are used instead; this avoids constructing objects of size G^H when G^H can exceed billions.

8 Reproducibility

Run the final simulation from the repository root:

```
Rscript scripts/sample_size/run_final_product_viroli_simulation.R.
```

The script is checkpointed through `comparison_results_checkpoint.csv`. Re-running the same command with `RESUME_EXISTING=TRUE` skips completed scenario-method pairs. The main output directory is

```
results/full/final_cross_ifeval_product_viroli/.
```

The final driver records the full simulation setting on every row, including the random seed scheme, penalties, mixture priors, convergence flags, elapsed time, and diagnostics. This makes the raw CSV the first place to audit any surprising result.